

SANYA HEDONG, China

WMO: 599485

Lat: 18.2508N

Lon: 109.5077E

Elev: 7

StdP: 101.24

Time Zone: 8.00 (E08)

Period: 94-08

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	16.3	17.4	7.4	6.4	19.6	9.9	7.6	19.4	4.6	23.6	4.3	23.6	1.4	40	0.282

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6	4.4	33.2	26.6	32.7	26.6	32.2	26.6	28.1	31.3	27.7	31.0	27.5	30.8	2.3	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
27.2	23.0	30.3	26.9	22.6	30.1	26.6	22.2	29.9	90.3	31.5	89.0	31.1	87.8	30.9	29.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
4.8	4.1	3.5	DB	13.7	34.4	1.9	0.8	12.4	35.0	11.3	35.4	10.2	35.9	8.8	36.4	(4)
			WB	10.0	28.5	2.2	0.3	8.5	28.8	7.2	29.0	6.0	29.2	4.4	29.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	26.9	22.9	23.6	25.6	28.0	29.6	29.2	28.9	28.3	27.7	25.9	23.7		
		DBStd	2.81	1.89	2.14	1.99	1.33	1.28	1.21	1.19	1.01	1.13	1.24	1.74	2.26	(7)
		HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0		
		HDD18.3	2	1	1	0	0	0	0	0	0	0	0	1		
		CDD10.0	6158	399	379	485	541	590	587	594	586	549	549	476	423	(10)
		CDD18.3	3119	142	147	227	291	332	337	336	327	299	291	226	167	(11)
		CDH23.3	31801	556	698	1723	3179	4119	4398	4216	3996	3368	2979	1761	807	(12)
		CDH26.7	11285	44	83	360	1093	1748	2036	1796	1579	1156	899	386	103	(13)
(14)	Wind	WSAvg	1.6	1.5	1.5	1.5	1.5	1.6	1.6	1.7	1.6	1.7	1.7	1.7	1.7	(14)
(15)	Precipitation	PrecAvg	1248	6	5	9	34	108	189	153	204	259	226	44	7	(15)
		PrecMax	1988	35	27	52	98	446	413	268	487	631	556	300	31	(16)
		PrecMin	674	0	0	0	0	10	6	18	39	47	12	0	0	(17)
		PrecStd	343	8	6	15	28	98	92	76	84	134	153	58	8	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4% DB	28.7	29.3	31.2	32.7	33.7	34.0	33.2	33.1	33.2	32.6	31.7	30.1		
		0.4% MCWB	22.3	22.7	24.2	25.4	26.5	26.9	26.9	26.8	25.8	24.9	24.6	23.6		
		2% DB	27.8	28.5	30.1	32.0	32.9	33.2	32.6	32.4	32.3	31.8	30.6	28.8		
		2% MCWB	21.9	22.6	24.2	25.5	26.6	27.2	26.9	26.8	25.8	24.8	23.9	22.7		
		5% DB	27.1	27.8	29.4	31.4	32.3	32.5	32.2	31.8	31.6	31.2	29.8	27.8		
		5% MCWB	21.4	22.3	23.9	25.4	26.6	27.2	26.9	26.8	25.8	24.8	23.4	21.9		
		10% DB	26.3	27.0	28.7	30.8	31.7	32.0	31.7	31.3	31.0	30.5	28.9	26.9		
		10% MCWB	21.0	21.8	23.6	25.3	26.5	27.1	26.9	26.7	25.9	24.6	23.0	21.3		
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4% WB	24.0	24.6	25.4	27.2	28.2	28.7	28.2	28.2	27.5	26.7	25.7	24.8		
		0.4% MCDB	26.6	27.3	28.9	30.5	31.6	31.9	31.4	31.2	30.8	30.6	29.6	28.3		
		2% WB	22.8	23.9	25.0	26.7	27.7	28.2	27.9	27.7	27.0	26.2	25.2	24.0		
		2% MCDB	26.7	26.8	28.5	30.0	31.2	31.5	31.0	30.8	30.4	30.0	28.7	26.9		
		5% WB	22.2	23.2	24.6	26.3	27.4	27.9	27.6	27.4	26.7	25.8	24.7	23.3		
		5% MCDB	25.8	26.4	28.1	29.8	30.8	31.1	30.8	30.5	30.0	29.4	28.1	26.4		
		10% WB	21.6	22.6	24.2	25.9	27.1	27.6	27.3	27.1	26.3	25.5	24.2	22.3		
		10% MCDB	25.1	25.9	27.7	29.5	30.5	30.9	30.5	30.2	29.6	28.9	27.6	26.0		
(35)	Mean Daily Temperature Range	MDBR	6.2	5.7	5.3	5.1	4.9	4.4	4.3	4.4	5.0	5.6	6.0	5.9		
		5% DB MCDBR	6.7	6.4	5.6	5.6	5.5	4.9	4.8	5.3	6.1	6.4	6.8	6.5		
		5% DB MCWBR	2.7	2.4	1.9	1.7	1.8	1.8	1.7	1.9	2.2	2.4	2.6	2.8		
		5% WB MCDBR	6.1	5.1	4.8	4.7	4.8	4.2	4.3	4.6	4.9	5.3	5.3	5.6		
		5% WB MCWBR	2.7	2.2	1.9	1.8	1.9	1.8	1.9	2.1	2.2	2.2	2.1	2.4		
(40)	Clear-Sky Solar Irradiance	taub	0.507	0.525	0.627	0.591	0.483	0.470	0.451	0.475	0.531	0.570	0.514	0.533		
		taud	2.022	2.013	1.791	1.932	2.285	2.336	2.387	2.302	2.106	1.946	2.074	1.961		
		Ebn at Noon	770	782	718	748	823	826	843	829	782	733	759	729		
		Edh at Noon	164	174	224	195	136	128	122	133	161	183	155	170		
(44)	All-Sky Solar Radiation	RadAvg	3.69	4.31	4.69	5.49	5.78	5.42	5.34	5.09	4.78	4.49	4.03	3.47		
		RadStd	0.46	0.60	0.35	0.43	0.39	0.45	0.51	0.36	0.49	0.54	0.33	0.35		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	+0.71	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (5 neighbors)	+0.26	N/A	N/A	+0.32	N/A	N/A	N/A	N/A	+90	+83

Nomenclature: See separate page